

Lab 8

Mixer & Preamp Assembly

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ECEN 299

Objectives

Objectives of this lab were to end up with our assembled board, specifically with the separate volume controls for the audio and microphone input while ensuring the daughterboard conforms to design specifications as required by the assignment.

Procedure

Equipment and supplies

- Oscilloscope
- Function Generator
- Soldering Iron and soldering materials
- Microscope
- Speakers
- Lab Mixer Board

Procedure

For this session, what we did first was of course assemble the Mixer and Preamp board; soldering components as specified to my design as made previously. After this assembly, we then used the function generator and the oscilloscope to analyze the board. First we checked the Pre-Amp, finding that it is withing an acceptable tolerance of what we were expecting. These values can be found for reference in Table 1 below.

Description	Vin(actual)	Vout(actual)	Av(actual)	Av(simulation) From lab 4
Pre-Amp Total	40mV	10.2V	250	274

Table 1 Actual (measured) results for pre-amplifier stages and total.

We then checked back to the table below which was from Lab 4 to compare to our actual values for the Pre-Amp against our simulated values. Note how there is some difference, but the values are within an acceptable tolerance to perform as required. Table 2 has our actual data values recorded in it from our analysis of the board, the gain and the error checked against the simulation values.

Description	Vin	Vout(calculated)	Av(calculated) Vout/Vin	Vout(simulated)	Av(simulated)
Pre-Amp, Stage 1	Vmin=40mV	Vout(s1)= -662mV	-16.56	-662mV	-16.55
Pre-Amp, Stage 2	Vout(s1)= -662mV	Vmout= 10.96V	-16.56	10.96V	-16.56
Pre-Amp, Total	Vmin= 40mV	10.96V	274	10.96V	274

Table from Lab 4

Amplifier	Frequency (Hz)	Vin(act)	Vout(act)	AdB(act)	AdB(cal) =20log(Av(cal))	Error = Adb(cal)-AdB(act)
Pre-Amp Get Av(cal) from lab4	20	40mV	10.5V	48.38	48.7	0.32
	200	40mV	10.5V	48.38	48.7	0.32
	2k	40mV	10.3V	48.22	48.7	0.48
	20k	40mV	10.2V	48.13	48.7	0.57
Mixer Left Channel	20	2V	4.0V	6.02	6	0.02
	200	2V	4.0V	6.02	6	0.02
	2k	2V	3.92V	5.85	6	0.15
	20k	2V	4.0V	6.02	6	0.02
Mixer Right Channel	20	2V	4.04V	6.11	6	0.11
	200	2V	4.04V	6.11	6	0.11
	2k	2V	4.2V	6.44	6	0.44
	20k	2V	4.08V	6.19	6	0.19

Table 2 Actual (measure) gains at various frequencies.

The data found in Table 3 shows our error values. By finding the error in it we can determine how well our board is likely to perform for our requirements.

Amplifier	Error (Emin)	Error(Emax)	Frequency Response Error Emax-Emin
Pre-Amp	0.32	0.57	0.25
Mixer (L)	.02	0.15	0.13
Mixer (R)	0.11	0.44	0.33

Table 3 – Frequency Response Error Calculation

Below with figures 1,2 and 3 we can see in real time values from the oscilloscope. These images depict the waveforms we saw as we took measurements of our data throughout the lab as we made our analysis of the different channels and circuits.

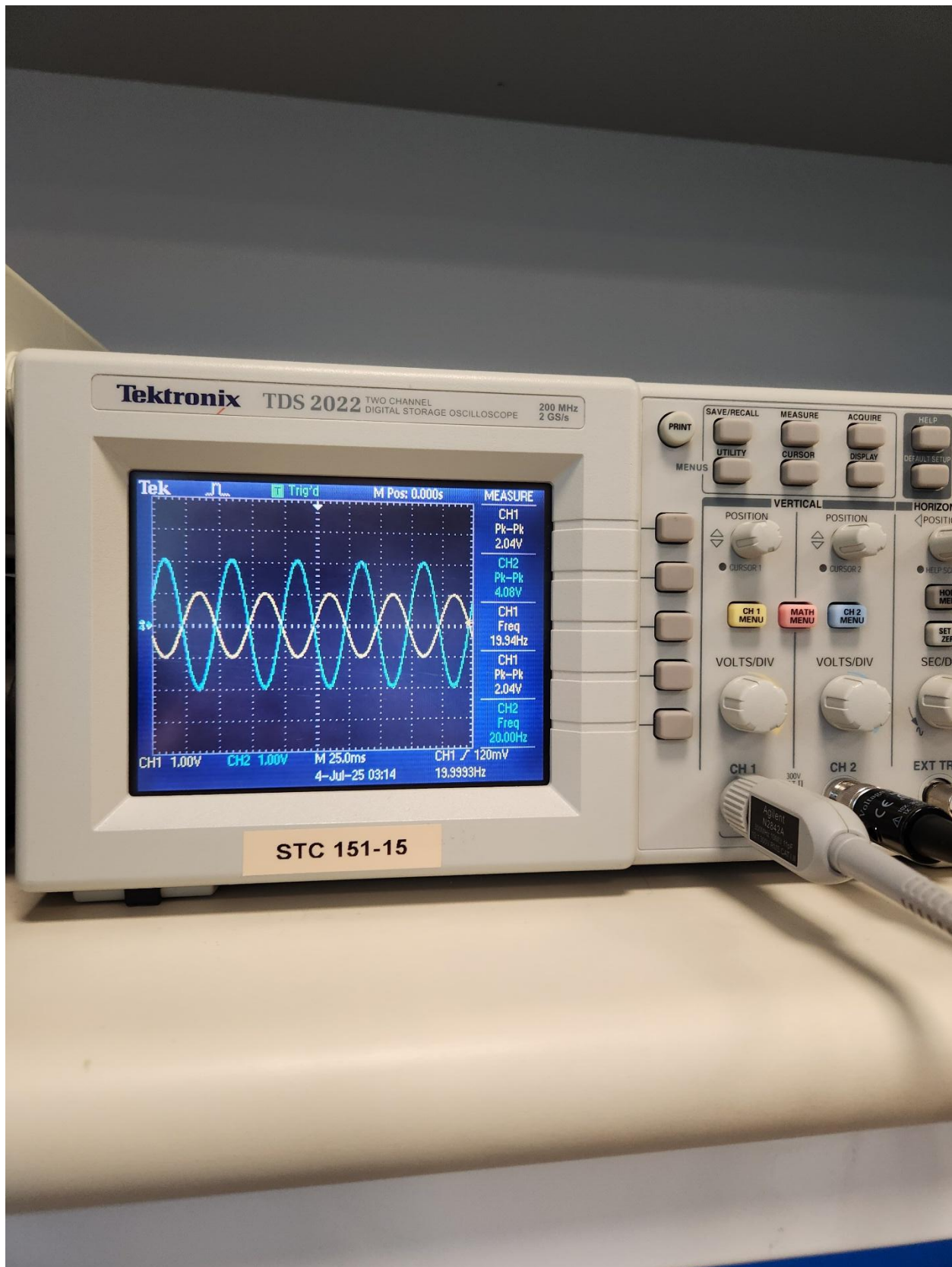


Figure 1: Mixer Right Channel (20kHz)

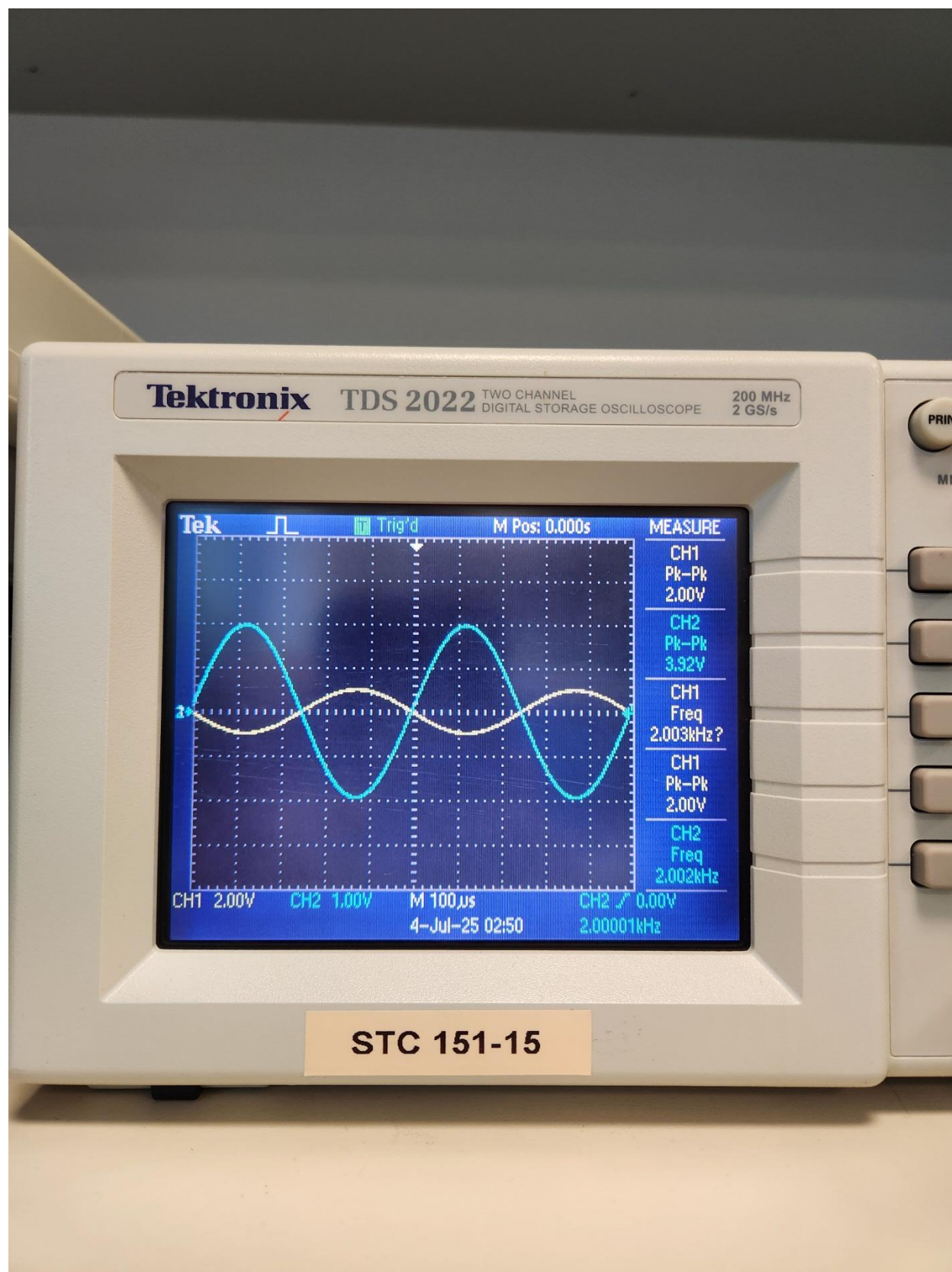


Figure 2: Mixer Left Channel (2kHz)

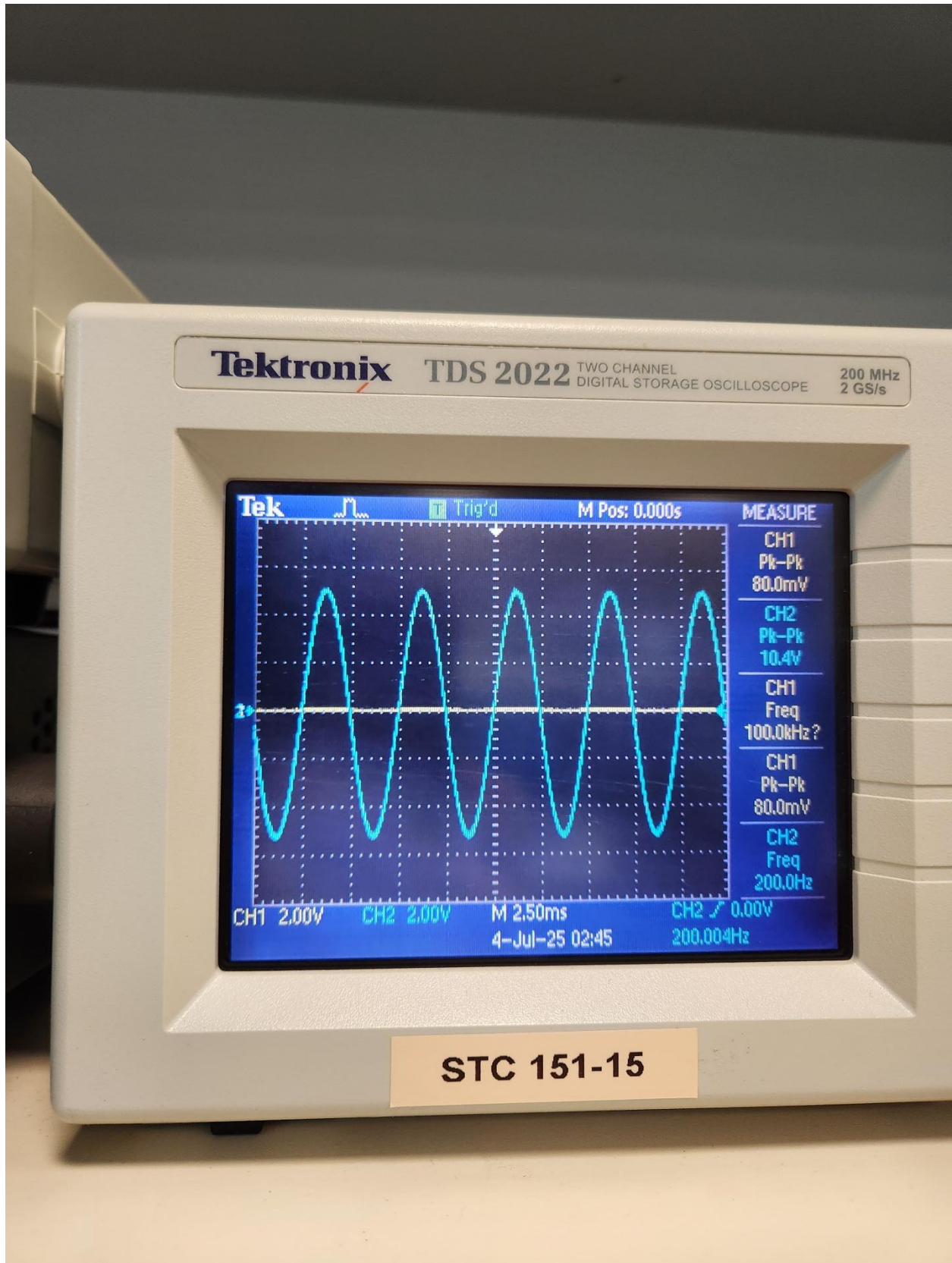


Figure 3: Pre-Amp (200Hz)

Conclusion

This lab session and project was not too difficult. This time some of the ideas were easier to complete now that I had spent some time doing more of these steps for the lab sessions. I am having better understanding of how the oscilloscope functions, how to troubleshoot it and to more readily look for things that may seem off from what I expect; the same goes for the function generator and some of the other lab equipment. There was some difficulty with some of the components as with how small they are it can be difficult to make sure the ends of resistors and capacitors are actually in contact with the PCB pads; I found using a DMM to be quite useful in ensuring connectivity for parts of the PCB and the soldered components. While not necessarily needed, I did also use the DMM to verify the resistance of the resistors to make sure they were what I was needing for the board. While I am still new to soldering, I believe I am beginning to find new strategies and practices that are helping me be more effective and efficient with the process of assembling these boards with their required components.